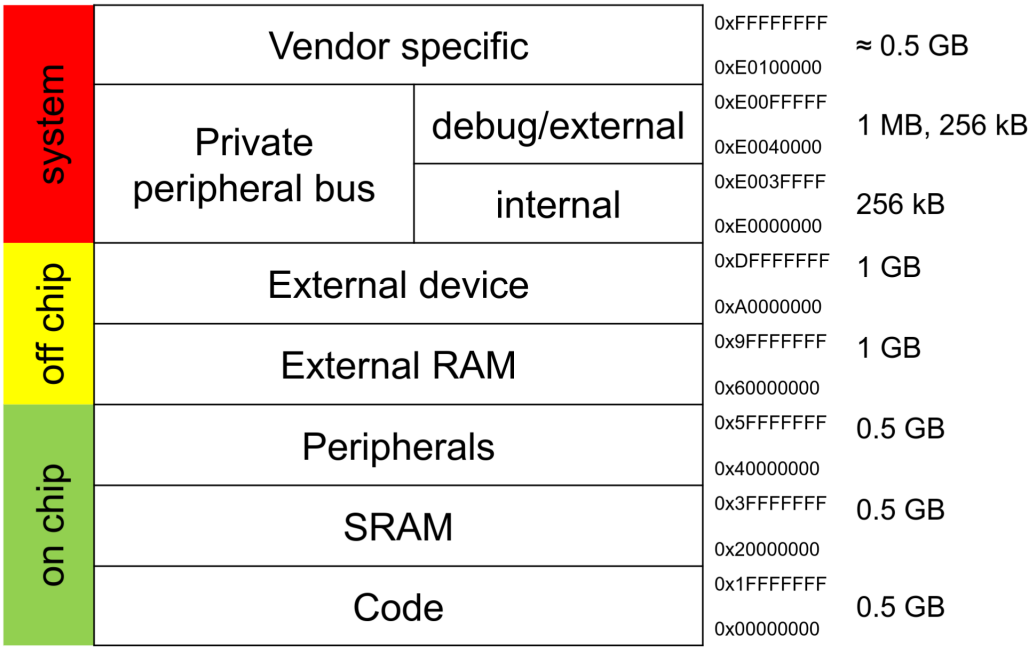
1 Lesson 13-10-26

1.1 Cortex-M3 memory map

Let's look at the Cortex-M3 memory map.



We see a bunch of significant sections:

- The first region, up to `0x1FFFFFFF`, is dedicated to code (on **FLASH** memory);
- We then have **SRAM** (for data, stack and heap regions), up to `0x3FFFFFFF`;
- Lastly we have **on-chip** *peripherals*, ranging up to `0x5FFFFFFF`;
- The region from `0x60000000` to `0xDFFFFFFF` is equally split between **external RAM** and **external devices**.
- Lstly, the region from `0xE0000000` is system-reserved, dedicated to the private peripheral bus and vendor specific data / registers.

Let's see a table of the memory region access characteristics:

Region	Bufferable	Cacheable	Executable
Code	yes	write through	yes
SRAM	yes	write back	yes
Peripherals	no	no	no
External RAM	no	write back	yes
External device	no	no	no
System	no	no	no